

Learning to Learn without Forgetting by Maximizing Transfer and Minimizing Interference

Motivation. Continual learning over non-stationary distributions remains a major challenge for neural networks. Prior work has framed the problem primarily through the lens of catastrophic forgetting (interference) or the stability-plasticity dilemma. However, these views consider only backward-in-time effects: how current learning degrades past knowledge.

This paper argues that a complete picture must be *temporally symmetric*, accounting for both backward interference and forward transfer simultaneously.

The Transfer-Interference Trade-off. The authors define operational measures of transfer and interference between two examples (x_i, y_i) and (x_j, y_j) via gradient alignment:

- **Transfer** occurs when $\frac{\partial L(x_i, y_i)}{\partial \theta} \cdot \frac{\partial L(x_j, y_j)}{\partial \theta} > 0$, meaning learning one example improves the other.
- **Interference** occurs when this dot product is negative, meaning learning one example harms the other.

Weight sharing between examples creates both transfer and interference potential. The key insight is that we must consider this trade-off in both the forward and backward temporal directions, not just backward as in the classical stability-plasticity framing.

Ideal Objective. To maximize transfer and minimize interference, the paper proposes the following idealized objective:

$$\theta = \arg \min_{\theta} \mathbb{E}_{[(x_i, y_i), (x_j, y_j)] \sim D} \left[L(x_i, y_i) + L(x_j, y_j) - \alpha \frac{\partial L(x_i, y_i)}{\partial \theta} \cdot \frac{\partial L(x_j, y_j)}{\partial \theta} \right].$$

The gradient dot-product regularizer directly encourages the network to share parameters where gradient directions align and keep them separate where gradients conflict.

Meta-Experience Replay (MER). Computing second derivatives of the ideal objective is expensive, so the authors approximate it using first-order meta-learning (Reptile) combined with experience replay. The resulting algorithm, **Meta-Experience Replay (MER)**, works as follows:

- (a) Maintain an experience replay buffer M updated via reservoir sampling.

- (b) At each time step, draw s batches of $k-1$ replay samples, each interleaved with the current example.
- (c) Process each batch with inner SGD steps, applying a within-batch Reptile meta-update after each batch.
- (d) Apply an across-batch Reptile meta-update to the main parameters.
- (e) Add the current example to the replay buffer.

This procedure approximately optimizes the transfer-interference objective with minimal computational overhead beyond standard experience replay.

Key Design Choices. Several properties make MER effective:

- **Reservoir sampling** ensures the buffer approximates the stationary distribution over all examples seen so far.
- **Prioritizing the current example** by interleaving it with every replay batch ensures the model can learn the present while consolidating past knowledge.
- **SGD batch size of 1** provides fine-grained second derivative information at the example level.

Experimental Results. MER is evaluated on supervised continual learning benchmarks and reinforcement learning settings:

- **MNIST Rotations/Permutations (20 tasks):** MER outperforms GEM, EWC, and online baselines in retained accuracy, showing superior balance between transfer and interference.
- **Many Permutations & Omniglot:** In more non-stationary settings (100 tasks, 50 alphabets), MER’s advantage over baselines grows substantially, nearly quadrupling baseline performance on Omniglot.
- **Buffer sensitivity:** MER’s benefits increase as the buffer shrinks, providing $>10\%$ gains over GEM at the smallest buffer sizes.
- **Continual RL (Catcher, Flappy Bird):** DQN-MER retains knowledge across task switches while standard DQN forgets catastrophically.

Gradient dot-product analysis confirms that MER produces significantly more positive gradient alignment than standard experience replay.

Limitations. MER requires maintaining an experience replay buffer, which may be infeasible at very large scales. The Reptile-based meta-learning adds computational overhead proportional to the number of replay batches. The theoretical analysis relies on first-order approximations of the ideal objective.

Takeaway. This paper reframes continual learning as solving the transfer-interference trade-off in a temporally symmetric manner. MER provides a practical, computationally efficient algorithm that combines experience replay with meta-learning to promote gradient alignment, consistently outperforming prior continual learning methods across supervised and reinforcement learning settings.